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Adventitial Fibroblasts and PI16+ Universal Fibroblasts Across Organs

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Background

Adventitial fibroblasts are part of a broader fibroblast family that builds and maintains connective tissue but is now known to do much more than matrix production. In blood vessels, the adventitia is no longer seen as a passive outer coat: it is a complex, active compartment where fibroblasts help sense stress, organize inflammation, and shape vessel remodeling. (12 sources)

Fibroblasts are the main mesenchymal cells of connective tissues and are classically understood as cells that make, assemble, and remodel extracellular matrix, including collagen and elastin, thereby helping set tissue structure and mechanics (Rajasekar et al., 2021). More recent work has expanded this view: fibroblasts also support immune surveillance, inflammation, blood vessel function, tissue-specific niches, and repair, and they show major functional diversity across organs and microenvironments (Kirk et al., 2021) (Rajasekar et al., 2021) (Sumbal et al., 2025). A key point for this review is that fibroblasts are still hard to define with a single universal marker set, and both classic morphology-based definitions and commonly used markers are imperfect (An et al., 2015) (Rajasekar et al., 2021) (Kirk et al., 2021).

Within this broader fibroblast landscape, the vascular adventitia has become recognized as a notable and highly active outer vessel-wall compartment rather than a simple supporting layer (An et al., 2015) (Stenmark et al., 2010) (Majesky et al., 2011) (Stenmark et al., 2012). The adventitia contains fibroblasts as its most abundant cell type, but also immune cells, progenitor cells, vasa vasorum endothelial cells, pericytes, and nerves, making it the most complex vessel-wall compartment in many descriptions (Stenmark et al., 2010) (Stenmark et al., 2012). This matters because adventitial fibroblasts are often among the first resident vascular cells to respond to injury or stress such as hypoxia, overdistension, or infection, after which they can proliferate, change phenotype, produce matrix, release cytokines, chemokines, growth factors, reactive oxygen species, and metalloproteinases, and help drive inflammation, vasa vasorum expansion, and remodeling from the “outside in” (An et al., 2015) (Stenmark et al._1, 2012) (Stenmark et al., 2010) (Stenmark et al., 2012).

This updated background also fits a larger concept in fibroblast biology: fibroblast identity is strongly shaped by anatomical position and local environment (Rajasekar et al., 2021) (Foote et al., 2019) (Rinn et al., 2006) (Shaw et al., 2020). Across the body, fibroblasts retain site-linked transcriptional programs, and single-cell studies now routinely split fibroblasts into anatomically distinct subtypes within the same organ, including adventitial fibroblasts in the lung (Rajasekar et al., 2021) (Rinn et al., 2006) (Liu et al., 2020). That broader context is important for understanding later sections: “adventitial fibroblast” names a location-linked stromal program around vessels, but that program can include both conserved features across organs and tissue-specific specializations.

Evidence

(Rajasekar et al., 2021)

[DNA Methylation of Fibroblast Phenotypes and Contributions to Lung Fibrosis]

> *"Fibroblasts synthesize and integrate structural proteins including collagen and elastin into the extracellular matrix (ECM) of mesenchymal tissues (Lynch et al., 2018). They are an integral part of connective tissue and play a crucial role in developing and modulating the structural framework of tissues by acting as the primary source of ECM. Furthermore, they incorporate mechanical properties to the ECM while dynamically modulating the architecture (Shaw et al., 2020)"*

> *".They were defined by their spindle-shaped morphology and characterized by expression of vimentin, procollagen Iα2 and fibroblast specific protein-1 (FSP1) (Lynch et al., 2018)(Rinn et al., 2006). However, these markers are not fibroblast specific, and a precise definition of the fibroblast remains elusive. Increasingly fibroblast heterogeneity is recognized across developmental stages, tissue of origin and microenvironment (Lynch et al., 2018), where they display an extensive variation in morphology, proliferation, function, molecular secretion (cytokines and ECM proteins) and molecular markers (Rinn et al., 2006)(Fries et al., 1994)(Jordana et al., 1988)(Ko et al., 1977)(Derdak et al., 1992). This heterogeneity is likely linked to their inherent plasticity and ability to specialize in different tissues (Shaw et al., 2020). Comparing transcriptomes of fibroblasts from the trachea, lung, abdomen, scalp, upper gingiva and soft palate (Foote et al., 2019) identified distinct gene expression profiles by body location"*

> *".Subsequently, fibroblasts with distinct gene expression profiles were defined at distinct anatomical locations in the lung; peribronchial cells in the conducting airway wall, adventitial fibroblasts around the bronchovascular bundles and alveolar fibroblasts embedded in alveolar regions of the lung [17]. However, across all single cell datasets, including healthy and diverse lung disease tissue, between 6 and 11 mesenchymal populations have been identified to date (Liu et al., 2020)."*

(Kirk et al., 2021)

[Fibroblast Memory in Development, Homeostasis and Disease]

> *"Today, fibroblasts are increasingly recognised for their contribution to immune surveillance and inflammation, blood vessel function, cancer progression, and maintenance of tissue specific structures (e.g., hair follicles) and stem cell niches (e.g., bone marrow, synovium). Recent advances in single cell transcriptomics have revealed that fibroblasts display significant functional heterogeneity, with fibroblast functions varying by their anatomical location and microenvironment (Shaw et al., 2020). There is ongoing work to generate a single cell atlas of fibroblast diversity across the whole organism in mice and humans in order to understand fibroblast development, heterogeneity, and disease (Buechler et al., 2021)"*

> *".Fibroblasts are the most common cells of connective tissues, but are a poorly defined population, with limited specific, universal markers [3]."*

(Sumbal et al., 2025)

[Contractile fibroblasts form a transient niche for the branching mammary epithelium]

> *"Recent studies employing single-cell transcriptomics suggest that fibroblasts represent a heterogeneous cell population with distinct cell types or cell states specialized for different functions, presenting inter-organ conserved as well as divergent behaviour and functions 4,5 ."*

(An et al., 2015)

[Characterization and Functions of Vascular Adventitial Fibroblast Subpopulations]

> *"Experimental data indicate that the vascular adventitia is a key regulator of vessel wall structure and function in health and disease (Wang et al., 2010)(Stenmark et al., 2012)(Majesky et al., 2011)[40][41]. It is considered that the vascular adventitia is the most complex compartment in the vessel wall structure and comprises mainly of adventitial fibroblasts (Stenmark et al., 2010). It is also clear that as the sensor, adventitial fibroblasts are first to be activated in response to vascular injury or environmental stress, and then undergoes phenotypic changes (Stenmark et al., 2012)(Sorrell et al., 2009)(Smith et al., 1997)(Yoshimura et al., 2006)(Phan, 2003)"*

> *".Beyond these morphologic and functional descriptions of adventitial fibroblasts, no reliable and specific cell marker for the fibroblast has been found to date. It is difficult with current tools to identify precisely fibroblasts in vivo and in vitro."*

(Stenmark et al., 2010)

[The Adventitia: Essential Role in Pulmonary Vascular Remodeling]

> *"A rapidly emerging concept is that the vascular adventitia acts as a biological processing center for the retrieval, integration, storage, and release of key regulators of vessel wall function. It is the most complex compartment of the vessel wall and comprises a variety of cells including fibroblasts, immunomodulatory cells, resident progenitor cells, vasa vasorum endothelial cells, and adrenergic nerves. In response to vascular stress or injury, resident adventitial cells are often the first to be activated and reprogrammed to then influence tone and structure of the vessel wall. Experimental data indicate that the adventitial fibroblast, the*

most abundant cellular constituent of adventitia, is a critical regulator of vascular wall function. In response to vascular stresses such as overdistension, hypoxia, or infection, the adventitial fibroblast is activated and undergoes phenotypic changes that include proliferation, differentiation, and production of extracellular matrix proteins and adhesion molecules, release of reactive oxygen species, chemokines, cytokines, growth factors, and metalloproteinases that, collectively, affect medial smooth muscle cell tone and growth directly and that stimulate recruitment and retention of circulating inflammatory and progenitor cells to the vessel wall. Resident dendritic cells also participate in "sensing" vascular stress and actively communicate with fibroblasts and progenitor cells to simulate repair processes that involve expansion of the vasa vasorum, which acts as a conduit for further delivery of inflammatory/progenitor cells. This review presents the current evidence demonstrating that the adventitia acts as a key regulator of pulmonary vascular wall function and structure from the "outside in." © 2011 American Physiological Society. Compr Physiol 1:141-161, 2011."

(Majesky et al., 2011)

The Adventitia: A Dynamic Interface Containing Resident Progenitor Cells

"Conventional views of the tunica adventitia as a poorly organized layer of vessel wall composed of fibroblasts, connective tissue and perivascular nerves are undergoing revision. Recent studies suggest that the adventitia has properties of a stem/progenitor cell niche in the artery wall that may be poised to respond to arterial injury. It is also a major site of immune surveillance and inflammatory cell trafficking, and harbors a dynamic microvasculature, the vasa vasorum, that maintains the medial layer and provides an important gateway for macrophage and leukocyte migration into the intima. In addition, the adventitia is in contact with tissue that surrounds the vessel and may actively participate in exchange of signals and cells between the vessel wall and the tissue in which it resides. This brief review highlights recent advances in our understanding of the adventitia and its resident progenitor cells and discusses progress toward an integrated view of adventitial function in vascular development, repair and disease."

(Stenmark et al., 2012)

The Adventitia: Essential Regulator of Vascular Wall Structure and Function

"The vascular adventitia acts as a biological processing center for the retrieval, integration, storage, and release of key regulators of vessel wall function. It is the most complex compartment of the vessel wall and is comprised of a variety of cells including fibroblasts, immunomodulatory cells (dendritic and macrophages), progenitor cells, vasa vasorum endothelial cells and pericytes, and adrenergic nerves. In response to vascular stress or injury, resident adventitial cells are often the first to be activated and re-programmed to then influence tone and structure of the vessel wall, to initiate and perpetuate chronic vascular inflammation, and to act to stimulate expansion of the vasa vasorum, which can act as a conduit for continued inflammatory and progenitor cell delivery to the vessel wall. This review presents the current evidence demonstrating that the adventitia acts as a key regulator of vascular wall function and structure from the "outside-in.""

(Stenmark et al._1, 2012)

Targeting the adventitial microenvironment in pulmonary hypertension: A potential approach to therapy that considers epigenetic change

"Experimental data indicate that the adventitial compartment of blood vessels, in both the pulmonary and systemic circulations, like the connective tissue stroma in tissues throughout the body, is a critical regulator of vessel wall function in health and disease. It is clear that adventitial cells, and in particular the adventitial fibroblast, are activated early following vascular injury, and play essential roles in regulating vascular wall structure and function through production of chemokines, cytokines, growth factors, and reactive oxygen species (ROS). The recognition of the ability of these cells to generate and maintain inflammatory responses within the vessel wall provides insight into why vascular inflammatory responses, in certain situations, fail to resolve. It is also clear that the activated adventitial fibroblast plays an important role in regulating vasa vasorum growth, which can contribute to ongoing vascular remodeling by acting as a conduit for delivery of inflammatory and progenitor cells. These functions of the fibroblast clearly support the idea that targeting chemokine, cytokine, adhesion molecule, and growth factor production in activated fibroblasts could be helpful in abrogating vascular inflammatory responses and thus in ameliorating vascular disease. Further, the recent observations that fibroblasts in vascular and fibrotic diseases may maintain their activated state through epigenetic alterations in key inflammatory and pro-fibrotic genes suggests that current therapies used to treat pulmonary hypertension may not be sufficient to induce apoptosis or to inhibit key inflammatory signaling pathways in these fibroblasts. New therapies targeted at reversing changes in the acetylation or methylation status of key transcriptional networks may be needed. At present, therapies specifically targeting abnormalities of histone deacytelase (HDAC) activity in fibroblast-like cells appear to hold promise."

(Foote et al., 2019)

Tissue specific human fibroblast differential expression based on RNAsequencing analysis

"Physical forces, such as mechanical stress, are essential for tissue homeostasis and influence gene expression of cells. In particular, the fibroblast has demonstrated sensitivity to extracellular matrices with assumed adaptation upon various mechanical loads. The purpose of this study was to compare the vocal fold fibroblast genotype, known for its unique mechanically stressful tissue environment, with cellular counterparts at various other anatomic locales to identify differences in functional gene expression profiles. By using RNA-seq technology, we identified differentially expressed gene programs (DEseq2) among seven normal human fibroblast primary cell lines from healthy cadavers, which included: vocal fold, trachea, lung, abdomen, scalp, upper gingiva, and soft

palate. Unsupervised gene expression analysis yielded 6216 genes differentially expressed across all anatomic sites. Hierarchical cluster analysis revealed grouping based on anatomic site origin rather than donor, suggesting global fibroblast phenotype heterogeneity. Sex and age-related effects were negligible. Functional enrichment analyses based on separate post-hoc 2-group comparisons revealed several functional themes within the vocal fold fibroblast related to transcription factors for signaling pathways regulating pluripotency of stem cells and extracellular matrix components such as cell signaling, migration, proliferation, and differentiation potential. Human fibroblasts display a phenomenon of global topographic differentiation, which is maintained in isolation via in vitro assays. Epigenetic mechanical influences on vocal fold tissue may play a role in uniquely modelling and maintaining the local environmental cellular niche during homeostasis with vocal fold fibroblasts distinctly specialized related to their anatomic positional and developmental origins established during embryogenesis."

(Rinn et al., 2006)

Anatomic Demarcation by Positional Variation in Fibroblast Gene Expression Programs

"Fibroblasts are ubiquitous mesenchymal cells with many vital functions during development, tissue repair, and disease. Fibroblasts from different anatomic sites have distinct and characteristic gene expression patterns, but the principles that govern their molecular specialization are poorly understood. Spatial organization of cellular differentiation may be achieved by unique specification of each cell type; alternatively, organization may arise by cells interpreting their position along a coordinate system. Here we test these models by analyzing the genome-wide gene expression profiles of primary fibroblast populations from 43 unique anatomical sites spanning the human body. Large-scale differences in the gene expression programs were related to three anatomic divisions: anterior-posterior (rostral-caudal), proximal-distal, and dermal versus nondermal. A set of 337 genes that varied according to these positional divisions was able to group all 47 samples by their anatomic sites of origin. Genes involved in pattern formation, cell-cell signaling, and matrix remodeling were enriched among this minimal set of positional identifier genes. Many important features of the embryonic pattern of HOX gene expression were retained in fibroblasts and were confirmed both in vitro and in vivo. Together, these findings suggest that site-specific variations in fibroblast gene expression programs are not idiosyncratic but rather are systematically related to their positional identities relative to major anatomic axes."

(Shaw et al., 2020)

Dissecting Fibroblast Heterogeneity in Health and Fibrotic Disease

"Purpose of Review Fibroblasts, the major cell population in all connective tissues, are best known for their role in depositing and maintaining the extracellular matrix. Recently, numerous specialised functions have been discovered revealing unpredicted fibroblast heterogeneity. We will discuss this heterogeneity, from its origins in development to alterations in fibrotic disease conditions. Recent Findings Advances in lineage tracing and single-cell transcriptional profiling techniques have revealed impressive diversity amongst fibroblasts in a range of organ systems including the skin, lung, kidney and heart. However, there are major challenges in assimilating the findings and understanding their functional significance. Certain fibroblast subsets can make specific contributions to healthy tissue functioning and to fibrotic disease processes; thus, therapeutic manipulation of particular subsets could be clinically beneficial. Summary Here we propose that four key variables determine a fibroblast's phenotype underpinning their enormous heterogeneity: tissue status, regional features, microenvironment and cell state. We review these in different organ systems, highlighting the importance of understanding the divergent fibroblast properties and underlying mechanisms in tissue fibrosis."

(Liu et al., 2020)

Definition and Signatures of Lung Fibroblast Populations in Development and Fibrosis in Mice and Men

"The heterogeneity of fibroblasts in the murine and human lung during homeostasis and disease is increasingly recognized. It remains unclear if the different phenotypes identified to date are characteristic of unique subpopulations with unique progenitors or whether they arise by a process of differentiation from common precursors. Our understanding of this ubiquitous cell type is limited by an absence of well validated, specific, markers with which to identify each cell type and a clear consensus on the distinct populations present in the lung. Here we describe single cell RNA sequencing (scRNA-seq) analysis on mesenchymal cells from the murine lung throughout embryonic (E) development (E9.5 – 17.5), at post-natal day (P1 – 15), as well as in the adult and the aged murine lungs before and after bleomycin-induced fibrosis. We carried out complementary scRNA-seq on human lung tissue from a P1 lung, a month 21 lung and lung tissue from healthy donors and patients with idiopathic pulmonary fibrosis (IPF). The murine and human data were supplemented with publicly available scRNA-seq datasets. We consistently identified lipofibroblasts, myofibroblasts, pericytes, mesothelial cells and smooth muscle cells. In addition, we identified a novel population delineated by Ebf1 (early B-cell factor 1) expression and an intermediate subtype. Comparative analysis with human mesenchymal cells revealed homologous mesenchymal subpopulations with remarkably conserved transcriptomic signatures. Comparative analysis of changes in gene expression in the fibroblast subpopulations from age matched non-fibrotic and fibrotic lungs in the mouse and human demonstrates that many of these subsets contribute to matrix gene expression in fibrotic conditions. Subtype selective transcription factors were identified and putative divergence of the clusters during development were delineated. Prospective isolation of these fibroblast subpopulations, localization of signature gene markers, and lineage-tracing each cluster are under way in the laboratory. This analysis will enhance our understanding of fibroblast heterogeneity in homeostasis and fibrotic disease conditions."

Structure, location, and baseline functions of adventitial fibroblasts

Adventitial fibroblasts are fibroblast-like stromal cells that sit in the outer vessel wall, just outside smooth muscle and around larger vessels. At baseline they help build and maintain the adventitial extracellular matrix, provide mechanical support, and organize local signaling, immune surveillance, repair, and remodeling responses. (8 sources)

Structurally, adventitial fibroblasts occupy the adventitia, the outermost connective tissue layer of the vessel wall, where they surround the media and form the interface between the vessel and nearby tissue (Cautivo et al., 2020) (Majesky, 2015). In the lung, spatial transcriptomic and imaging work places adventitial fibroblasts around larger blood vessels, where they form diffuse layers outside the tighter concentric rings of endothelial cells, pericytes, and vascular smooth muscle cells (He et al., 2022). This matches the broader view that large-vessel-associated stromal cells are a distinct perivascular compartment, different from capillary pericytes, parenchymal stromal cells, and capsular stromal cells (Cautivo et al., 2020).

Their baseline structural role is to make and maintain the adventitial extracellular matrix, especially fibrillar collagen-rich connective tissue that gives the vessel outer-wall support (Mackay et al., 2022) (Crnkovic et al., 2025). In large elastic arteries such as the aorta, the adventitial matrix can act as an external mechanical scaffold that helps limit vessel expansion, and this compartment also contains the vasa vasorum that support the outer wall and can serve as routes for immune-cell traffic (Cantalupo et al., 2026). More generally, fibroblasts use collagen, elastin, fibronectin, proteoglycans, and glycosaminoglycans to build and preserve connective tissue, and adventitial fibroblasts apply these core fibroblast functions to the vessel outer wall (Koczkowska et al., 2025).

Baseline function goes beyond passive support. Adventitial stromal cells are described as fibroblast-like cells that respond to hypoxia, mechanical stretch, cytokines, and growth factors, and in doing so organize, remodel, and repair the adventitial environment (Cautivo et al., 2020) (Majesky, 2015). In developing lung, their transcriptomes are enriched for extracellular matrix organization and signaling pathways such as BMP, TGF-β, and WNT, which fits a role in maintaining the perivascular niche and supporting vessel-associated tissue architecture (He et al., 2022). They also sit within a broader adventitial cuff that can coordinate immune crosstalk and local tissue immunity around vessels (He et al., 2022) (Dahlgren et al., 2019).

When stress or injury occurs, these same cells can shift into activated repair states. Across vascular beds, adventitial fibroblasts can proliferate, become myofibroblast-like, migrate locally, and increase secretion of collagen, cytokines, and other remodeling signals, thereby linking matrix repair to inflammation (Mackay et al., 2022) (Koczkowska et al., 2025). Disease-focused studies in the pulmonary artery and aorta reinforce that this is not just a theoretical capacity: adventitial fibroblasts can become strongly inflammatory and matrix-remodeling cells during remodeling while still remaining recognizably within the fibroblast lineage (Crnkovic et al., 2025) (Cantalupo et al., 2026).

Evidence

(Cautivo et al., 2020)

Immune outposts in the adventitia: One foot in sea and one on shore

> *"Here we group SCs by microanatomic location: capillary-associated (pericytes), body cavity-associated (capsular SCs), parenchymal-associated (parenchymal SCs) and large vessel-associated (adventitial stromal cells, ASCs). ASCs are predominantly fibroblast like cells that are responsive to various local stimuli, including hypoxia, mechanical stretch, cytokines, and growth factors, organizing, remodeling, and repairing the adventitial environment (Majesky, 2015). ASCs from multiple tissues express platelet-derived growth factor receptor α (PDGFRα) and podoplanin (Gp38), with variable combinations of Sca-1, Gli-1, CD34, and IL-33 [10]."*

(Majesky, 2015)

Adventitia and Perivascular Cells

> *"The past few years have seen significant advances in our understanding of the multiple and dynamic roles of the adventitia and its companion perivascular tissues for vessel wall homeostasis and disease. It is now becoming clear that signals originating from within the adventitia and from perivascular cells play essential roles in regulation of vascular development, physiology, artery wall remodeling, immune surveillance, and vascular disease.1–4 The adventitia is in contact with and completely surrounds the media and is the interface between the vessel wall and its neighboring tissues. It contains many different interacting cell types including fibroblasts, microvascular endothelium, nerves, resident macrophages, T cells, B cells, mast cells, and dendritic cells.5–9 The*

adventitia is also home to resident vascular progenitor cells whose formation and maintenance depend, in part, on sonic hedgehog signaling.1,10–12

Perivascular cells are in close contact with the adventitia, particularly for the aorta and coronary arteries. Perivascular tissue includes adipocytes, lymphatic vessels, perivascular nerves, and stromal cells exhibiting mesenchymal stem cell–like properties.3,13 The adventitia and periadventitial cells function in concert. They are linked by microvessels, nerves, and migratory cells to regulate vascular physiology, homeostasis, structural remodeling, and exert major influences on the progression or regression of vascular disease. Crosstalk between intima, media, and adventitia further links the adventitia-periadventitial unit to the rest of the vessel wall. Much of the work advancing our concepts of the adventitia and periadventitial tissues has been published in ATVB , and some of those key studies are discussed in this highlights article. In the following summary, areas that have attracted the most interest over the past 2 years in ATVB are reviewed including the multiple roles of perivascular adipose tissue (PVAT) on control of vascular physiology and remodeling, adventitial progenitor …"

(He et al., 2022)

[A human fetal lung cell atlas uncovers proximal-distal gradients of differentiation and key regulators of epithelial fates]()

"In the oldest lungs sequenced, there were three distinct fibroblasts: adventitial (SFRP2 + , PI16 +), airway (AGTR2 + , S100A4 +), and alveolar (WNT2 + , FGFR4 +) with distinct locations (Figures 4A, 4B, and 4D-4G)"

".he three major fibroblast types in 15-22 pcw lungs expressed high levels of genes associated with ECM organization but had distinct gene expression patterns and spatial localization. Adventitial fibroblasts (SFRP2 + , PI16 +) surrounded the larger blood vessels (Figure 4D). They formed diffuse layers of cells surrounding the tightly packed concentric rings of ECs, pericytes, and vSMCs (Figures 4E and S7H). Adventitial fibroblasts were enriched in genes associated with ECM organization and signaling, including BMP, TGFb, and WNT (Figures 4J, 4K, and S7J), consistent with described roles providing structural support to the perivascular region. (Dahlgren et al., 2019)"

(Mackay et al., 2022)

[Adventitial Fibroblasts in Aortic Aneurysm: Unraveling Pathogenic Contributions to Vascular Disease]()

"Adventitial fibroblasts are one of the key constituents of the aortic wall, and they play an essential role in maintaining vessel structure and function"

".Adventitial fibroblasts facilitate the production of extracellular matrix (ECM), providing structural integrity. However, during biomechanical stress and/or injury, adventitial fibroblasts can be activated into myofibroblasts, which move to the site of injury and secrete collagen and cytokines, thereby enhancing the inflammatory response"

".Adventitial fibroblasts play an essential role in the structure and function of the vascular wall"

".Furthermore, technical challenges surrounding the research of adventitial fibroblasts are significant, which emphasizes the need for proper techniques for maintenance and markers for the identification of fibroblasts."

(Crnkovic et al., 2025)

[Adventitial fibroblasts direct smooth muscle cell-state transition in pulmonary vascular disease]()

"Classical view of vascular smooth muscle cells as resident cells mediating vasoactive responses and adventitial fibroblasts as structural support cells producing fibrillar collagen has been appended by single-cell transcriptomics studies in humans. Single-cell resolution has helped to define the tissuespecific heterogeneity within and delineate the distinctions of vascular resident cell populations to other smooth muscle and fibroblast populations (Crnkovic et al., 2022)(Muhl et al., 2022)(Muhl et al., 2020)(Travaglini et al., 2019)"

".Fate mapping studies in animal models of systemic and pulmonary vascular disease have shown that the majority of vascular smooth muscle cells and adventitial fibroblasts retain their lineage identity during vascular remodeling (Biasin et al., 2020)(Bordenave et al., 2020)(Sheikh et al., 2015)(Steffes et al., 2020)(Wirka et al., 2019)(Worssam et al., 2022). However, a subset of these cells exhibits the acquisition or loss of specific markers, such as smooth muscle actin (Chu et al., 2024)(Schwartz, 2018)(Short et al., 2004). Nevertheless, diseased cells consistently demonstrate altered expression profiles compared to their healthy counterparts (Crnkovic et al., 2022)(Wirka et al., 2019)(Worssam et al., 2022)(Zhang et al., 2024)."

(Cantalupo et al., 2026)

[Mechanotransduction in Marfan Syndrome and Related Aortic Disorders: Insights from Transcriptomic Analyses]()

"The adventitial ECM provides structural support to the aortic wall and has been proposed to function as an external mechanical scaffold that limits vessel expansion. Consistent with this concept, fibrillin-1 is not restricted to elastic fiber-associated microfibrils within the media but also forms independent microfibrillar networks in connective tissues, including the adventitia, suggesting that pathogenic variants affecting fibrillin biology may directly influence adventitial structure and function (Lindeman et al., 2009)"

".Recent advances in sc/snRNA-seq transcriptomics have further expanded our understanding of adventitial contributions to aortic disease. Fibroblasts exhibit substantial phenotypic heterogeneity rather than representing a homogeneous structural population, and undergo marked transcriptional reprogramming during aneurysm development. Studies in Loeys-Dietz syndrome mouse models identified distinct fibroblast populations characterized by ECM remodeling, inflammatory signaling, and stress-response programs, indicating active participation in disease progression rather than a purely supportive role (Jana et al., 2025)(Dalal et al., 2025). These observations are consistent with emerging findings from human and experimental aneurysm datasets demonstrating that fibroblasts contribute to ECM turnover, intercellular signaling, and immune cell recruitment"

".In addition to fibroblasts, the adventitial compartment contains the vasa vasorum, a specialized microvascular network that supplies oxygen and nutrients to the outer regions of the aortic wall and may also serve as a route for immune-cell trafficking."

(Koczkowska et al., 2025)

Identifying differentiation markers between dermal fibroblasts and adipose-derived mesenchymal stromal cells (AD-MSCs) in human visceral and subcutaneous tissues using single-cell transcriptomics

"Fibroblasts play a crucial role in the formation of connective tissue in the human body. Their primary function is to produce and maintain the extracellular matrix (ECM) by secreting essential components such as collagen, elastin, fibronectin, proteoglycans and glycosaminoglycans (Solé-Boldo et al., 2020). During tissue regeneration, fibroblasts migrate to injury sites, facilitate ECM production, and regulate inflammation through cytokine and growth factor release (Talbott et al., 2022)"

".our scRNA-seq analysis elucidates the transcriptional landscape of AD-MSCs and fibroblasts with unprecedented resolution, highlighting both the population-specific markers and the intrapopulation heterogeneity"

".we pinpointed 30 genes exhibiting the most significant variations in expression between AD-MSCs and fibroblasts. These genes are associated with biological processes such as tissue remodeling, cell movement, and activation in response to external stimuli. Among them, MMP1, MMP3, S100A4, CXCL1, PI16, IGFBP5, COMP were further validated using qPCR, clearly demonstrating their potential to differentiate between AD-MSCs and fibroblasts."

(Dahlgren et al., 2019)

Adventitial cuffs: regional hubs for tissue immunity

"Inflammation must be effective while limiting excessive tissue damage. To walk this line, immune functions are grossly compartmentalized by innate cells that act locally and adaptive cells that function systemically. But what about the myriad tissue-resident immune cells that are critical to this balancing act and lie on a spectrum of innate and adaptive immunity? We propose that mammalian perivascular adventitial 'cuffs' are conserved sites in multiple organs, enriched for these tissue-resident lymphocytes and dendritic cells, as well as lymphatics, nerves, and subsets of specialized stromal cells. Here, we argue that these boundary sites integrate diverse tissue signals to regulate the movement of immune cells and interstitial fluid, facilitate immune crosstalk, and ultimately act to coordinate regional tissue immunity."

Cell type markers and distinction from related stromal/mural cells

Adventitial fibroblasts are still best defined by a mix of location and transcriptomic profile rather than by any single perfect marker. The most useful current distinction is "ECM-rich, mural-marker-negative fibroblasts in the adventitia," with PI16 emerging as a strong and relatively specific marker in several organs. (14 sources)

- **Location is still part of the definition.**
  In healthy arteries, adventitial fibroblasts are commonly identified by their position in the outer vessel wall, outside the smooth muscle layer and separated from it by the external elastic lamina, because no single marker cleanly captures all fibroblasts. This is why morphology, exclusion of other lineages, and vessel-wall anatomy remain part of the practical definition. (Kuret et al., 2021) (Kuwabara et al., 2017)
- **Classic "fibroblast markers" are useful but not specific.**
  Common markers such as **vimentin, PDGFRA, FSP1/S100A4, discoidin-domain receptor, and prolyl-4-hydroxylase** are all imperfect because they can be expressed by other cell types and are not present in all fibroblasts. In human aorta, even widely used markers such as **FSP1/S100A4, PDGFRA, COL1A1, and GLI1** were reported to be diffuse across several non-fibroblast populations. (Kuret et al., 2021) (Zhao et al., 2025) (Kramann et al., 2014)
- **A practical positive marker set is ECM-rich fibroblast genes.**
  Across vascular studies, adventitial fibroblasts are often recognized by high expression of fibrillar and matrix-associated genes such as **COL1A1, COL1A2, COL3A1, DCN, LUM**, and related collagen-binding

proteins, together with reduced expression of vascular smooth muscle contractile genes. In mouse aorta, fibroblasts were defined by **PDGFRA** plus collagens and matrix genes, with lower **Myh11** and **Cnn1** than VSMCs. (Kuret et al., 2021) (Kalluri et al., 2019) (Lendahl et al., 2022)

- **The main transcriptomic distinction from mural cells is absence of canonical mural markers.**
  A key modern separation is between fibroblasts and mural cells such as pericytes and smooth muscle cells. Adventitial fibroblasts express matrix genes, including **COL1A1/COL3A1, DCN, and PI16**, but lack classic mural markers such as **CSPG4/NG2, MCAM/CD146, RGS5, and ACTA2**. This helps distinguish them from pericytes and vSMCs even when histology alone is ambiguous. (Lendahl et al., 2022) (Muhl et al., 2020)
- **Distinction from smooth muscle cells can be subtle in transcriptomic space.**
  In some datasets, adventitial fibroblasts form a continuum with SMCs, which likely reflects shared mesenchymal programs and close physical location. Even so, the fibroblast population remains identifiable because it lacks contractile SMC markers and instead shows high expression of extracellular matrix components and modifiers. (Zhao et al., 2025)
- **PI16 is one of the strongest current markers for vascular adventitial fibroblasts.**
  In both broad fibroblast reviews and vascular single-cell studies, **PI16** repeatedly appears as a distinguishing marker of adventitial fibroblasts. Human aortic work identified **PI16, SFRP2, and MFAP5** as especially specific for the adventitial fibroblast population, while lung and multi-organ studies also place **PI16** in adventitial/perivascular fibroblasts. (Lendahl et al., 2022) (Zhao et al., 2025)
- **Other useful vascular fibroblast markers include DPEP1 and PDGFRA.**
  Recent arterial work validated **PDGFRA** and **DPEP1** by immunohistochemistry and flow cytometry across human and mouse aorta, carotid, and femoral arteries as fibroblast-enriched markers, while showing that traditional markers like **CD90** and **vimentin** also labeled **transgelin+** vascular smooth muscle cells. This makes **DPEP1** and **PDGFRA** more useful than older broad markers in arterial wall studies, though still best interpreted in combination. (Kuijk et al., 2023)
- **CD90 marks a rare adventitial stromal subset, not all adventitial fibroblasts.**
  In human medium- and large-sized arteries, **CD90** marked a rare adventitial population that co-expressed MSC-associated markers such as **PDGFRα, CD44, CD73, and CD105**. However, those accompanying markers were broadly expressed by other cells, so CD90 here appears to identify a specific **MSC-like adventitial subset** rather than the whole adventitial fibroblast compartment. (Michelis et al., 2018)
- **Sca-1/Ly6a is often used in mouse arterial adventitial fibroblasts, but it reflects a mouse-specific stromal state.**
  Mouse arterial studies note that fibroblasts in the adventitia often express **Sca-1/Ly6a**, and this has been linked to stromal plasticity and progenitor-like behavior. In lung single-cell datasets, an adventitial fibroblast population was defined as **Sca1+ Ly6C+ Cd9+** and also expressed **Pi16, Dpt, and Col14a1**, consistent with a vessel-associated fibroblast identity. (Kuijk et al., 2023) (Soucy et al., 2025)
- **Markers can also help separate adventitial fibroblasts from other fibroblast subtypes in the same organ.**
  In lung, adventitial fibroblasts can be separated from alveolar fibroblasts by marker combinations. Adventitial cells were **Sca1+ Ly6C+ Cd9+ Pi16+ Dpt+ Col14a1+**, whereas alveolar fibroblasts were **Pdgfra+ Col13a1+ Npnt+** and also expressed **Ces1d** and **Tcf21**. This shows that “fibroblast” is not one vessel-adjacent state but a set of location-linked stromal programs. (Soucy et al., 2025) (Tsukui et al., 2020) (Travaglini et al., 2019)
- **There is real heterogeneity within adventitial fibroblasts, so no marker list is complete.**
  Earlier work described morphologic subtypes such as **epithelioid-like** and **spindle-like** adventitial fibroblasts in rat aorta, with different responses to angiotensin II. Single-cell studies have since found multiple fibroblast or mesenchymal subclusters in aortic adventitia and trajectory-like states marked by genes such as **CD55, CXCL14, and LOX**, again arguing against a single uniform marker set. (Kuret et al.,

2021) (An et al., 2015) (Gu et al., 2019) (Kuijk et al., 2023)

- **Best current working definition.**
  Taken together, the most defensible current definition of adventitial fibroblasts is: **cells located in the vascular adventitia that express fibroblast/ECM genes such as COL1A1, COL3A1, DCN, LUM, PDGFRA, PI16, DPT, COL14A1, SFRP2, MFAP5, and sometimes DPEP1, while lacking canonical mural markers such as ACTA2, CSPG4/NG2, MCAM/CD146, RGS5, MYH11, and CNN1**. Spatial methods such as RNAscope, immunostaining, and spatial transcriptomics are often needed to confirm that a transcriptomic fibroblast cluster is truly adventitial. (Lendahl et al., 2022) (Zhao et al., 2025) (Kuijk et al., 2023) (Kuret et al., 2021)

Evidence

(Kuret et al., 2021)

The Role of Fibroblasts in Atherosclerosis Progression

> *"In a healthy artery, adventitial fibroblasts are found in a non-active, quiescent state, and are usually defined by their location in the vessel wall since they can be separated from the more generally recognized smooth muscle cell layer by an external elastic lamina (Kuwabara et al., 2017)(Michel et al., 2007). All currently used markers to identify fibroblasts, including vimentin, platelet derived growth factor receptor α (PDGFRα), fibroblast specific protein 1 (FSP1), discoidin-domain receptor, and prolyl-4-hydroxylase, are potentially problematic, as they are also expressed in other cell types and are not present in all fibroblasts (Chang et al., 2002)(Matsuyama et al., 2006)(Goodpaster et al., 2008). Therefore, to identify fibroblasts, investigators have to rely on the lack of markers for other cell lineages (e.g. non-lymphoid, non-endothelium, and non-epithelium), along with morphologic, functional, and biochemical characteristics (Chang et al., 2002)"*

> *".Adventitial fibroblasts show differences in morphology, size, function and activity in the healthy, as well as stressed conditions or disease states (Tinajero et al., 2019). For example, An et al. (An et al., 2015) found two major fibroblast subpopulations in the adventitia of rat thoracic aorta. The two populations were described as epithelioid-like cells and spindle-like cells, however only epithelioid-like fibroblasts were sensitive to stimulation with angiotensin II, a hormone involved in the development of hypertension and atherosclerosis [28]"*

> *".Das et al. (Das et al., 2002) concluded that numerous phenotypically and biochemically distinct fibroblast subpopulations can be found and only a selective increase in the number of resident fibroblast subpopulations with enhanced growth capability was observed under hypoxic conditions (Das et al., 2002)"*

> *".Kalluri et al. (Kalluri et al., 2019) explored the cellular atlas of healthy mice aortas using single cell RNA sequencing. They showed that fibroblasts represent approximately 33% of all aortic cells and were defined by higher expression of PDGFRα and collagens/collagen-binding proteins (e.g. Col1a1, Col1a2, Dcn, Lum) whereas the expression of VSMC-associated contractile proteins (e.g. Myh11, Cnn1) was reduced. These fibroblasts clustered into two subpopulations and are probably derived from the adventitia, however their exact location needs to be confirmed by immunohistochemistry or in-situ hybridization (Kalluri et al., 2019). Another study using the high resolution single cell analysis approach, was performed by Gu et al. (Gu et al., 2019) in aortic adventitial cells from wild type and apolipoprotein E-deficient (ApoE−/−) mice. They determined four heterogeneous mesenchymal populations with differential gene expression suggesting potential functions in ECM organization, immune regulation and bone formation."*

(Kuwabara et al., 2017)

Tracking Adventitial Fibroblast Contribution to Disease

> *"Cells present in the adventitia, or outermost layer of the blood vessel, contribute to the progression of vascular diseases, such as atherosclerosis, hypertension, and aortic dissection. The adventitial fibroblast of the aorta is the prototypic perivascular fibroblast, but the adventitia is composed of multiple distinct cell populations. Therefore, methods for uniquely identifying the fibroblast are critical for a better understanding of how these cells contribute to disease processes. A popular method for distinguishing adventitial cell types relies on the use of genetic tools in the mouse to trace and manipulate these cells. As reporter and Cre recombinase expressing mice are used more frequently in studies of vascular disease, it is important to outline the advantages and limitations of these genetic tools. The purpose of this review is to provide an overview of the various genetic tools available in the mouse for the study of resident adventitial fibroblasts."*

(Zhao et al., 2025)

A cell and transcriptome atlas of human arterial vasculature

> *"The second most abundant population of cells isolated in our dataset are adventitial fibroblasts, which form a continuum in the transcriptomic space with SMCs. This population of cells do not express contractile SMC markers (Figures 1C and 2A) but do express high levels of extracellular matrix components and modifiers (Figure 3A). Marker gene RNAscope studies for this*

population, along with spatial transcriptomics, localized this population exclusively to the adventitia (Figures 3B and 1D). Based on marker gene expression, we termed this population "vascular adventitial fibroblast" (AdvFib)"

".Commonly used canonical fibroblast marker genes, (Soliman et al., 2021)(Kramann et al., 2014) including fibroblast specific protein (FSP1/S100A4), PDGFRA, COL1A1, and GLI1, were expressed relatively diffusely in several different populations of non-fibroblast cells (Figure 3C). 4](Philippeos et al., 2018) Using scRNA-seq data, we additionally identified PI16, SFRP2, and MFAP5 as the most specific marker genes for this adventitial fibroblast population (Figures 3B-3D; Table S3). Of note, genes highly expressed specifically in the adventitial fibroblast population were enriched for genes known to cause genetic aortopathies, (Pedroza et al., 2020)(Renner et al., 2019)[48] including FBN1, TGFBR2, LOX, FBLN1, and MFAP5 (Figure 3E)"

".While it has been described that the adventitial fibroblast is a heterogeneous population, they appeared to be relatively homogenous in the transcriptomic space within the same vascular segments. There were additional ECs and other microvascular cells that are in the adventitia, which makes it more heterogeneous than other layers, but the fibroblasts themselves within each segment appeared to be mostly one large continuous population across a spectrum. However, there is significant heterogeneity across the adventitial fibroblasts across different vascular beds (Figure S3A)"

".Focusing on the transcriptomic heterogeneity across AdvFib, three distinct clusters highlighted by different biological processes can be found (Figures S3B and S3C). Pathway analysis revealed particular sensitivity of each of these clusters to different signaling pathway (Figure S3D). One cluster has enrichment in fibroblast growth factor and promigration signal (we termed "FGF-avid"), another has higher TGFB pathway and ossification signals (we termed "TGFBavid"), and the third has enrichment for BMP signals and microfilaments (we termed "BMP-avid") cluster (Figures S3C-S3E). The "TGFB-avid" fibroblasts are particularly enriched segments with the neural crest origin, whereas the "BMP-avid" cluster is particularly enriched from segments with the paraxial/ splenic mesodermal origin (Figure S3A)."

(Kramann et al., 2014)

Perivascular Gli1+ progenitors are key contributors to injury-induced organ fibrosis

"Summary Mesenchymal stem cells (MSCs) reside in the perivascular niche of many organs, including kidney, lung, liver, and heart, although their roles in these tissues are poorly understood. Here, we demonstrate that Gli1 marks perivascular MSC-like cells that substantially contribute to organ fibrosis. In vitro, Gli1+ cells express typical MSC markers, exhibit trilineage differentiation capacity, and possess colony-forming capacity, despite constituting a small fraction of the platelet-derived growth factor-β (PDGFRβ)+ cell population. Genetic lineage tracing analysis demonstrate that tissue-resident, but not circulating, Gli1+ cells proliferate following kidney, lung, liver, or heart injury to generate myofibroblasts. Genetic ablation of these cells substantially ameliorates kidney and heart fibrosis, and preserves ejection fraction in a model of induced heart failure. These findings implicate perivascular Gli1+ MSC-like cells as a major cellular origin of organ fibrosis and demonstrate these cells may be a relevant therapeutic target to prevent solid organ dysfunction following injury."

(Kalluri et al., 2019)

Single cell analysis of the normal mouse aorta reveals functionally distinct endothelial cell populations

"Background The cells that form the arterial wall contribute to multiple vascular diseases. The extent of cellular heterogeneity within these populations has not been fully characterized. Recent advances in single cell RNA-sequencing makes it possible to identify and characterize cellular subpopulations. Methods We validate a method for generating a droplet-based single cell atlas of gene expression in a normal blood vessel. Enzymatic dissociation of four whole mouse aortas was followed by single cell sequencing of over 10,000 cells. Results Clustering analysis of gene expression from aortic cells identified 10 populations of cells representing each of the main arterial cell types—fibroblasts, vascular smooth muscle cells (VSMCs), endothelial cells (ECs), and immune cells including monocytes, macrophages, and lymphocytes. The most significant cellular heterogeneity was seen in the 3 distinct EC populations. Gene set enrichment analysis of these EC subpopulations identified a lymphatic EC cluster and two other populations more specialized in lipoprotein handling, angiogenesis, and extracellular matrix production. These subpopulations persist and exhibit similar changes in gene expression in response to a Western diet. Immunofluorescence for Vcam1 and Cd36 demonstrates regional heterogeneity in EC populations throughout the aorta. Conclusions We present a comprehensive single cell atlas of all cells in the aorta. By integrating expression from over 1,900 genes per cell we are better able to characterize cellular heterogeneity compared with conventional approaches. Gene expression signatures identify cell subpopulations with vascular disease-relevant functions."

(Lendahl et al., 2022)

Identification, discrimination and heterogeneity of fibroblasts

"Adventitial fibroblasts are associated with larger blood vessels and are assumed to serve as structural support cells and to produce ECM components and growth factors important for vessel wall integrity and homeostasis. Histologically, fibroblasts in the adventitia can be difficult to distinguish from vSMCs and pericytes. Electron microscopy studies have, however, shown that fibroblasts in the adventitia exhibit morphological features common with fibroblasts in other organs, such as extensive rough endoplasmic reticulum and lack of a basement membrane64. Transcriptomically, adventitial fibroblasts are characterized by expression of collagens (for example, COL1A1 and COL3A1), the ECM glycoprotein decorin (DCN), and the peptidase inhibitor PI16, and they are distinct from

mural cells by lack of classic mural markers such as CSPG4/NG2, MCAM/CD146, RGS5 and ACTA2"

".PI16+ fibroblasts were found in the adventitia of arteries and veins in several organs and may represent a universal adventitial fibroblast population"

".In a cross-comparison of mouse fibroblasts isolated directly from different muscular organs and analyzed by scRNAseq, it was evident that differential expression of genes related to the ECM-the matrisome-is a major feature of inter-organ fibroblast transcriptomic heterogeneity (Muhl et al., 2020) (Fig. 2)."

(Muhl et al., 2020)

Single-cell analysis uncovers fibroblast heterogeneity and criteria for fibroblast and mural cell identification and discrimination

"Our data reveal common markers defining fibroblasts in all organs, but also an extensive fibroblast organotypicity, as well as interorgan heterogeneities. While no singular marker could discriminate all fibroblasts from all mural cells, we provide a 90-gene signature capable of doing so across all organs studied herein, as well as in other public datasets (Schaum et al., 2018)"

". Fibroblasts are important producers of ECM at many locations in the body"

"Our transcriptomic data reveal extensive inter-organ transcriptional heterogeneity among fibroblasts, differences that are particularly obvious regarding expression of genes in the matrisome (Naba et al., 2011) . The differences in matrisome gene expression were larger than for other gene/protein categories, indicating that fibroblasts specifically tailor the ECM in accordance with the organ-specific physiology (Bonnans et al., 2014)"

". the combination of cellular dispersion in UMAP and in situ mapping of anatomical localization of cells using relevant markers, as done herein, should provide physiologically meaningful annotations."

(Kuijk et al., 2023)

Human and murine fibroblast single-cell transcriptomics reveals fibroblast clusters are differentially affected by ageing and serum cholesterol

"Fibroblasts mostly reside in the adventitial layer of the arterial wall, accompanied by other mesenchymal cells [e.g. pericytes and smooth muscle cells (SMCs)], immune cells, and connective tissue. (Majesky et al., 2011) Mainly fibroblasts express the stem cell marker Sca-1/Ly6a, underpinning the potential of these cells to be reprogrammed into a diverse cell repertoire, supporting extensive plasticity. (Lee et al., 2016)(Xiao et al., 2018) Their functional role in fibrosis, inflammation, and angiogenesis in other organs (Shi et al., 1997)(Li et al., 2016) makes these cells an attractive candidate for therapeutic intervention in arterial pathologies, such as atherosclerosis and vascular ageing"

". scRNA-seq has been key in identifying pan-fibroblast-specific markers across the microvasculature in several major organs compared with mural cells (MCs) (consisting of pericytes and SMCs). 16"

"Murine aorta single-cell RNA-sequencing analysis of adventitial mesenchymal cells identified fibroblast-specific markers. Immunohistochemistry and flow cytometry validated platelet-derived growth factor receptor alpha (PDGFRA) and dipeptidase 1 (DPEP1) across human and murine aorta, carotid, and femoral arteries, whereas traditional markers such as the cluster of differentiation (CD)90 and vimentin also marked transgelin+ vascular smooth muscle cells. Next, pseudotime analysis showed multiple fibroblast clusters differentiating along trajectories. Three trajectories, marked by CD55 (Cd55+), Cxcl chemokine 14 (Cxcl14+), and lysyl oxidase (Lox+), were reproduced in an independent RNA-seq dataset. Gene ontology (GO) analysis showed divergent functional profiles of the three trajectories, related to vascular development, antigen presentation, and/or collagen fibril organization, respectively."

(Michelis et al., 2018)

CD90 Identifies Adventitial Mesenchymal Progenitor Cells in Adult Human Medium- and Large-Sized Arteries

"MSCs reportedly exist in a vascular niche occupying the outer adventitial layer. However, these cells have not been well characterized in vivo in medium- and large-sized arteries in humans"

"We identified that CD90 marks a rare adventitial population that co-expresses MSC markers including PDGFRα, CD44, CD73, and CD105. However, unlike CD90, these additional markers were widely expressed by other cells. Human adventitial CD90+ cells fulfilled standard MSC criteria, including plastic adherence, spindle morphology, passage ability, colony formation, and differentiation into adipocytes, osteoblasts, and chondrocytes. Phenotypic and transcriptomic profiling, as well as adoptive transfer experiments, revealed a potential role in vascular disease pathogenesis, with the transcriptomic disease signature of these cells being represented in an aortic regulatory gene network that is operative in atherosclerosis."

(Soucy et al., 2025)

Transcriptomic responses of lung mesenchymal cells during pneumonia

"Using genes that differentiate subsets recognized as adventitial or alveolar fibroblasts (Buechler et al., 2021)(Xie et al., 2018)

(Travaglini et al., 2019)55), as well as Louvain clustering of this dataset, this large butterfly cluster distinguishes adventitial Sca1 + Ly6C + Cd9 + cells (clusters 0, 1, 6, 12, and 16; Figure 3, C-E) and alveolar Pdgfra + Col13a1 + Npnt + (clusters 2, 3, 7, 8, 9, 10, and 17; Figure 3, F-H)"

".In addition to being Pdgfra + Col13a1 + Npnt + , the fibroblasts in the right-side clusters also expressed Ces1d and Tcf21 (Figure 3I and Supplemental Figure 4, A and B), which mark alveolar fibroblasts specifically (Tsukui et al., 2020). Consistent with their potential alveolar localization and function, these cells expressed Bmp3, Bmp4, and Vegfa (Figure 3I and Supplemental Figure 4, C-E), mesenchymal products essential to supporting alveolar epithelial cells and pulmonary capillary endothelial cells (Chung et al., 2018)"

".The Sca1 + Ly6C + Cd9 + cells expressed Pi16, Dpt, and Col14a1 (Figure 3I and Supplemental Figure 4, F-H), suggesting they are the adventitial fibroblasts found across multiple organs (56)."

(Tsukui et al., 2020)

Collagen-producing lung cell atlas identifies multiple subsets with distinct localization and relevance to fibrosis

"Collagen-producing cells maintain the complex architecture of the lung and drive pathologic scarring in pulmonary fibrosis. Here we perform single-cell RNA-sequencing to identify all collagen-producing cells in normal and fibrotic lungs. We characterize multiple collagen-producing subpopulations with distinct anatomical localizations in different compartments of murine lungs. One subpopulation, characterized by expression of Cthrc1 (collagen triple helix repeat containing 1), emerges in fibrotic lungs and expresses the highest levels of collagens. Single-cell RNA-sequencing of human lungs, including those from idiopathic pulmonary fibrosis and scleroderma patients, demonstrate similar heterogeneity and CTHRC1-expressing fibroblasts present uniquely in fibrotic lungs. Immunostaining and in situ hybridization show that these cells are concentrated within fibroblastic foci. We purify collagen-producing subpopulations and find disease-relevant phenotypes of Cthrc1-expressing fibroblasts in in vitro and adoptive transfer experiments. Our atlas of collagen-producing cells provides a roadmap for studying the roles of these unique populations in homeostasis and pathologic fibrosis. Collagen production by lung cells is critical to maintain organ architecture but can also drive pathological scarring. Here the authors perform single cell RNA sequencing of collagen-producing lung cells identifying a subset of pathologic fibroblasts characterized by Cthrc1 expression which are concentrated within fibroblastic foci in fibrotic lungs and show a pro-fibrotic phenotype."

(Travaglini et al., 2019)

A molecular cell atlas of the human lung from single cell RNA sequencing

"RNA in situ hybridization of the homologous mouse cells showed one population (cluster 30, marked by SPINT2, FGFR4, ITGA8) localized to the alveolus ("alveolar fibroblasts") (Fig. 2k) and the other (cluster 29, marked by SFRP2, PI16, SERPINF1) to vascular adventitia ("adventitial fibroblasts") (Fig. 2l). Both express genes involved in canonical fibroblast functions including extracellular matrix biosynthesis, cell adhesion, and angiogenesis regulators and other intercellular signals and modulators. However the specific genes for these functions often differ, for example adventitial fibroblasts expressed different Wnt pathway modulators (SFRP2) than those of alveolar fibroblasts (NKD1, RSPO1) (Fig. S2e). Each cluster also appears to have distinct functions"

"The expression profiles also suggest novel, shared immune functions including immune cell recruitment (IL1RL1, IL32, CXCL2, MHCII) and the complement system (C2, C3, C7, CFI, CFD, CFH, CFB) (Table S4)."

(An et al., 2015)

Characterization and Functions of Vascular Adventitial Fibroblast Subpopulations

"Experimental data indicate that the vascular adventitia is a key regulator of vessel wall structure and function in health and disease (Wang et al., 2010)(Stenmark et al., 2012)(Majesky et al., 2011)[40][41]. It is considered that the vascular adventitia is the most complex compartment in the vessel wall structure and comprises mainly of adventitial fibroblasts (Stenmark et al., 2010). It is also clear that as the sensor, adventitial fibroblasts are first to be activated in response to vascular injury or environmental stress, and then undergoes phenotypic changes (Stenmark et al., 2012)(Sorrell et al., 2009)(Smith et al., 1997)(Yoshimura et al., 2006)(Phan, 2003)"

".Beyond these morphologic and functional descriptions of adventitial fibroblasts, no reliable and specific cell marker for the fibroblast has been found to date. It is difficult with current tools to identify precisely fibroblasts in vivo and in vitro."

(Gu et al., 2019)

Adventitial Cell Atlas of wt (Wild Type) and ApoE (Apolipoprotein E)-Deficient Mice Defined by Single-Cell RNA Sequencing

"Supplemental Digital Content is available in the text."

Transcriptomic characterization of adventitial fibroblasts

Single-cell and spatial transcriptomic studies now define adventitial fibroblasts as a reproducible vessel-associated *fibroblast state rather than just a histologic location. Their core program combines extracellular-matrix and signaling genes with immune and complement-related modules, and in several datasets this state further splits into quiescent/structural versus immune-activated or niche-like subprograms. (11 sources)*

Transcriptomic work has made adventitial fibroblasts easier to recognize as a distinct fibroblast population across tissues. A notable early single-cell lung atlas separated fibroblasts into at least two location-linked states and used RNA in situ hybridization to place **SFRP2+ PI16+ SERPINF1+** cells in the **vascular adventitia**, distinct from **SPINT2+ FGFR4+ ITGA8+** alveolar fibroblasts in the alveolus. Both populations kept core fibroblast functions, but they differed in the specific genes they used for matrix, adhesion, angiogenesis, and signaling, including different WNT-pathway modulators such as **SFRP2** in adventitial fibroblasts versus **NKD1/RSPO1** in alveolar fibroblasts. The same study also pointed to shared but still important immune-facing features in adventitial fibroblasts, including genes linked to **immune cell recruitment** and **complement** pathways. (Travaglini et al., 2019)

Later lung studies refined this picture. In a developmental and injury-oriented mouse lung atlas, adventitial fibroblasts were marked by **Pi16, Clec3b, and Serpinf1** and showed enrichment for **ECM organization, complement regulation, and wound healing** programs. That same dataset also found a nearby fibroblast population marked by **Adh7** and **Bmp7**, enriched for **steroid metabolic processes**, and linked to adventitial cuffs, suggesting that even within vessel-adjacent fibroblasts there are separable transcriptomic niches rather than one uniform adventitial state. (Mayr et al., 2022) (Tsukui et al., 2020)

A recurring theme is that adventitial fibroblasts mix structural and immune-regulatory features. In pulmonary adventitial fibroblasts analyzed in relation to asthma genetics, reclustering resolved **19 transcriptionally distinct clusters**. These could be grouped into an **immune-activated antigen-presenting state** with **HLA-DRA** and **CD74**, enriched for **antigen processing and presentation, NF-κB signaling, and inflammatory response** pathways, and a more **matrix/vascular/metabolic state** enriched for **MFAP5, COL3A1**, vascular development, oxidative phosphorylation, and glutathione metabolism. Disease-associated variants were more strongly concentrated in the immune-activated group, which supports the idea that adventitial fibroblasts can shift between homeostatic structural roles and active immune-interacting states. (Zhou et al., 2025)

This split between structural and immune-niche programs also appears outside the lung. In human aortic aneurysm single-cell data, two fibroblast clusters located mainly in the **adventitia** expressed a familiar adventitial set including **PI16, PDGFRα, CCDC80, DPT,** and **CXCL12**, with a second cluster also showing **TNC** and **SERPINE1**. The authors interpreted these as resembling two broader fibroblast archetypes: **PI16+ adventitial fibroblasts**, described as relatively **structural and progenitor-like**, and **CXCL12+ stromal niche fibroblasts**, described as more **immune-interacting and paracrine** and colocalizing with lymphocytes and inflammatory macrophages. Importantly, both states remained spatially tied to the adventitia, suggesting that the adventitial compartment can contain more than one stable transcriptomic program. (Levy-Lambert et al., 2025) (Doring et al., 2014) (Gao et al., 2024) (Shaikh et al., 2024) (Steele et al., 2025)

Cross-tissue atlases strengthen the case that this is not a lung-only or aorta-only label. In reproductive tissues, adventitial fibroblasts emerged as a distinct transcriptomic compartment defined by **DPT, SFRP2, and C3**, forming a continuum from **PI16hi** states to **C7/COL15A1hi** states. Spatial mapping placed the **PI16hi** cells in subserosal or adventitial connective tissue and the **C7/COL15A1hi** states more broadly in interstitial and perivascular positions. The authors explicitly linked these states to previously described **universal fibroblast populations**, implying that adventitial fibroblasts are not just organ-specific annotations but part of a broader conserved fibroblast program with local variants. (Cohen et al., 2026) (Liu et al., 2025)

Taken together, transcriptomics supports a working model in which adventitial fibroblasts are identified by a conserved vessel-associated fibroblast core—often centered on **PI16, SFRP2, SERPINF1, DPT, CLEC3B, C3, PDGFRA,** and related ECM genes—plus superimposed modules for **complement**, **wound repair**, **CXCL12-mediated niche signaling**, or **HLA-DRA/CD74-associated antigen presentation**, depending on tissue and disease context. Spatial validation has been critical in these studies, because the transcriptomic state is reproducible but often blends into neighboring interstitial fibroblast states and can be missed if annotated only as generic stromal fibroblasts. (Travaglini et al., 2019) (Mayr et al., 2022) (Levy-Lambert et al., 2025) (Zhou et al., 2025) (Cohen et al., 2026)

Evidence

(Travaglini et al., 2019)

A molecular cell atlas of the human lung from single cell RNA sequencing

> *"RNA in situ hybridization of the homologous mouse cells showed one population (cluster 30, marked by SPINT2, FGFR4, ITGA8) localized to the alveolus ("alveolar fibroblasts") (Fig. 2k) and the other (cluster 29, marked by SFRP2, PI16, SERPINF1) to vascular adventitia ("adventitial fibroblasts") (Fig. 2l). Both express genes involved in canonical fibroblast functions including extracellular matrix biosynthesis, cell adhesion, and angiogenesis regulators and other intercellular signals and modulators. However the specific genes for these functions often differ, for example adventitial fibroblasts expressed different Wnt pathway modulators (SFRP2) than those of alveolar fibroblasts (NKD1, RSPO1) (Fig. S2e). Each cluster also appears to have distinct functions"*

> *"The expression profiles also suggest novel, shared immune functions including immune cell recruitment (IL1RL1, IL32, CXCL2, MHCII) and the complement system (C2, C3, C7, CFI, CFD, CFH, CFB) (Table S4)."*

(Mayr et al., 2022)

Autocrine Sfrp1 inhibits lung fibroblast invasion during transition to injury induced myofibroblasts

> *"Adventitial fibroblast marker genes included Pi16, Clec3b and Serpinf1 (Fig. 1b) and were enriched for genes involved in `ECM organization', `complement regulation' and `wound healing' (Fig. 1c). In addition, we identified a population of fibroblasts that featured expression of the All-trans-retinol dehydrogenase [NAD(+)] ADH7 and Bmp7, which we termed Adh7+ fibroblasts (Fig. 1b). The marker genes of these cells were interestingly enriched for `steroid metabolic processes' (Fig. 1c), and a recent study localized such cells to adventitial cuffs (Tsukui et al., 2020) ."*

(Tsukui et al., 2020)

Collagen-producing lung cell atlas identifies multiple subsets with distinct localization and relevance to fibrosis

> *"Collagen-producing cells maintain the complex architecture of the lung and drive pathologic scarring in pulmonary fibrosis. Here we perform single-cell RNA-sequencing to identify all collagen-producing cells in normal and fibrotic lungs. We characterize multiple collagen-producing subpopulations with distinct anatomical localizations in different compartments of murine lungs. One subpopulation, characterized by expression of Cthrc1 (collagen triple helix repeat containing 1), emerges in fibrotic lungs and expresses the highest levels of collagens. Single-cell RNA-sequencing of human lungs, including those from idiopathic pulmonary fibrosis and scleroderma patients, demonstrate similar heterogeneity and CTHRC1-expressing fibroblasts present uniquely in fibrotic lungs. Immunostaining and in situ hybridization show that these cells are concentrated within fibroblastic foci. We purify collagen-producing subpopulations and find disease-relevant phenotypes of Cthrc1-expressing fibroblasts in in vitro and adoptive transfer experiments. Our atlas of collagen-producing cells provides a roadmap for studying the roles of these unique populations in homeostasis and pathologic fibrosis. Collagen production by lung cells is critical to maintain organ architecture but can also drive pathological scarring. Here the authors perform single cell RNA sequencing of collagen-producing lung cells identifying a subset of pathologic fibroblasts characterized by Cthrc1 expression which are concentrated within fibroblastic foci in fibrotic lungs and show a pro-fibrotic phenotype."*

(Zhou et al., 2025)

Single-Cell Mapping of Genetic Risk Across Ten Respiratory Diseases

> *"Pulmonary artery adventitial fibroblasts were shown to direct the behavior of immune cells and serving as a homing place for the immune cells during vascular remodeling (Chelladurai et al., 2022). We observed substantial heterogeneity in the genetic association between adventitial fibroblasts and asthma (Figure 6a). Reclustering analysis further resolved this cell type into 19 transcriptionally distinct clusters (Figure 6b), revealing marked differences in disease association strength across subpopulations. Based on single-cell-level association patterns, adventitial fibroblasts could be stratified into two major functional groups (Figure 6c). Group A (clusters 4/9/11/12/15) was characterized by strong expression of HLA-DRA, CD74, and was enriched for pathways related to antigen processing and presentation, NF-κB signaling, and inflammatory responses (Figure 6d,e), with 2.1% of cells in these clusters significantly associated with the disease (Supplementary Table S2). In contrast, Group B (clusters 0/3/8) was enriched for extracellular-matrix and vascular/metabolic genes, including MFAP5, and COL3A1, with pathway enrichment for vascular development, oxidative phosphorylation, and glutathione metabolism (Figure 6d,f), with only 0.06% of cells in these clusters significantly associated with the disease (Supplementary Table S2). This expression profile suggests a tissue-remodeling/ metabolic-adaptation state, consistent with fibroblast contributions to extracellular matrix turnover and airway structural changes in chronic asthma (Weitoft et al., 2014). Notably, Group A exhibited significantly stronger genetic association with asthma in scDRS analysis than Group B, indicating that asthma risk variants are preferentially localized to fibroblast subpopulations with an immune-activated, antigen-presenting phenotype. This highlights the dual roles of fibroblasts in asthma pathogenesis: not only as structural mediators of remodeling but also as active participants in immune regulation."*

(Levy-Lambert et al., 2025)

Spatial Transcriptomics Reveals CXCL12+ Fibroblasts as Central Immune Organizers through CXCR4 Signaling in Abdominal Aortic Aneurysm

> *"Cluster 10 (universal adventitial fibroblast 1), the largest fibroblast cluster in controls (3.3% of cells), located predominantly in the adventitia, expressed PI16, (Steele et al., 2025)(Gao et al., 2024) PDGFRα, 57 CCDC80, (Gao et al., 2024) DPT, (Buechler et al., 2021) and CXCL12 (Döring et al., 2014) (Supplementary Tables 3 and 4). In AAA, these fibroblasts comprised 6.3% of cells and were also located in the adventitia (Supplementary Table 4, Figure 3D and E). Cluster 18 (universal adventitial fibroblast 2), located predominantly in the adventitia (Figure 3D and E), also expressed those genes, along with CXCL12, TNC, (Golledge et al., 2011) DPT, (Buechler et al., 2021) and SERPINE1 (Shaikh et al., 2024) (Figure 2G, Supplementary Table 3). Cluster 18 represented 1.4% of cells in AAAs but only 0.0007% of cells in control aortas. These clusters resemble two universal fibroblast archetypes: PI16⁺ adventitial fibroblasts (structural, progenitor-like), which are quiescent in uninjured tissues, 88 and CXCL12⁺ stromal niche fibroblasts (immune-interacting and paracrine hubs), 84 which colocalize with lymphocytes and inflammatory macrophage (Figure 3A) and likely help shape the tissue microenvironment through CXCL12 signaling to immune cells."*

(Doring et al., 2014)

The CXCL12/CXCR4 chemokine ligand/receptor axis in cardiovascular disease

> *"The chemokine receptor CXCR4 and its ligand CXCL12 play an important homeostatic function by mediating the homing of progenitor cells in the bone marrow and regulating their mobilization into peripheral tissues upon injury or stress. Although the CXCL12/CXCR4 interaction has long been regarded as a monogamous relation, the identification of the pro-inflammatory chemokine macrophage migration inhibitory factor (MIF) as an important second ligand for CXCR4, and of CXCR7 as an alternative receptor for CXCL12, has undermined this interpretation and has considerably complicated the understanding of CXCL12/CXCR4 signaling and associated biological functions. This review aims to provide insight into the current concept of the CXCL12/CXCR4 axis in myocardial infarction (MI) and its underlying pathologies such as atherosclerosis and injury-induced vascular restenosis. It will discuss main findings from in vitro studies, animal experiments and large-scale genome-wide association studies. The importance of the CXCL12/CXCR4 axis in progenitor cell homing and mobilization will be addressed, as will be the function of CXCR4 in different cell types involved in atherosclerosis. Finally, a potential translation of current knowledge on CXCR4 into future therapeutical application will be discussed."*

(Gao et al., 2024)

Cross-tissue human fibroblast atlas reveals myofibroblast subtypes with distinct roles in immune modulation.

> *"Fibroblasts, known for their functional diversity, play crucial roles in inflammation and cancer. In this study, we conduct comprehensive single-cell RNA sequencing analyses on fibroblast cells from 517 human samples, spanning 11 tissue types and diverse pathological states. We identify distinct fibroblast subpopulations with universal and tissue-specific characteristics. Pathological conditions lead to significant shifts in fibroblast compositions, including the expansion of immune-modulating fibroblasts during inflammation and tissue-remodeling myofibroblasts in cancer. Within the myofibroblast category, we identify four transcriptionally distinct subpopulations originating from different developmental origins, with LRRC15+ myofibroblasts displaying terminally differentiated features. Both LRRC15+ and MMP1+ myofibroblasts demonstrate pro-tumor potential that contribute to the immune-excluded and immune-suppressive tumor microenvironments (TMEs), whereas PI16+ fibroblasts show potential anti-tumor functions in adjacent non-cancerous regions. Fibroblast-subtype compositions define patient subtypes with distinct clinical outcomes. This study advances our understanding of fibroblast biology and suggests potential therapeutic strategies for targeting specific fibroblast subsets in cancer treatment."*

(Shaikh et al., 2024)

A signaling pathway map of plasminogen activator inhibitor-1 (PAI-1/SERPINE-1): a review of an innovative frontier in molecular aging and cellular senescence

> *"Plasminogen activator inhibitor-1 (PAI-1) is a vital regulator of the fibrinolytic mechanism and has been intricately involved in various physiological and clinical processes, including cancer, thrombosis, and wound healing. The PAI-1 signaling pathway is multifaceted, encompassing numerous signaling molecules and nodes. Recent studies have revealed a novel contribution of PAI-1 during cellular senescence. This review introduces a pathway resource detailing the signaling network events mediated by PAI-1. The literature curated on the PAI-1 system was manually compiled from various published studies, our analysis presents a signaling pathway network of PAI-1, which includes various events like enzyme catalysis, molecular association, gene regulation, protein expression, and protein translocation. This signaling network aims to provide a detailed analysis of the existing understanding of the PAI-1 signaling pathway in the context of cellular senescence across various research models. By developing this pathway, we aspire to deepen our understanding of aging and senescence research, ultimately contributing to the pursuit of effective therapeutic approaches for these complex chronic diseases."*

(Steele et al., 2025)

A single-cell and spatial genomics atlas of human skin fibroblasts reveals shared disease-related fibroblast subtypes across tissues

"Fibroblasts sculpt the architecture and cellular microenvironments of various tissues. Here we constructed a spatially resolved atlas of human skin fibroblasts from healthy skin and 23 skin diseases, with comparison to 14 cross-tissue diseases. We define six major skin fibroblast subtypes in health and three that are disease-specific. We characterize two fibroblast subtypes further as they are conserved across tissues and are immune-related. The first, F3: fibroblastic reticular cell-like fibroblast (CCL19+CD74+HLA-DRA+), is a fibroblastic reticular cell-like subtype that is predicted to maintain the superficial perivascular immune niche. The second, F6: inflammatory myofibroblasts (IL11+MMP1+CXCL8+IL7R+), characterizes early human skin wounds, inflammatory diseases with scarring risk and cancer. F6: inflammatory myofibroblasts were predicted to recruit neutrophils, monocytes and B cells across multiple human tissues. Our study provides a harmonized nomenclature for skin fibroblasts in health and disease, contextualized with cross-tissue findings and clinical skin disease profiles. Steele et al. use single-cell RNA sequencing and spatial transcriptomics to provide an atlas of adult human skin fibroblasts in healthy and diseased human skin."

(Cohen et al., 2026)

[An integrated multimodal pan-organ atlas of the female reproductive system across the lifespan contextualises gynaecological pathologies]

"In contrast, adventitial fibroblasts emerged as a distinct transcriptomics compartment that had been largely overlooked in prior atlases, likely owing to their scarcity and their annotation as generic stromal fibroblasts (Extended Data Fig. 3a and Supplementary Note 2.1). Pan-reproductive adventitial fibroblasts are defined by expression of DPT, SFRP2 and C3, forming a transcriptional continuum from PI16 hi (named "AdvFib PI16hi") to C7/COL15A1 hi states (labeled "AdvFib PI16low" and "Adv Fib (Intr)"; Figure 2a,c), consistent with the universal fibroblast populations previously described (Buechler et al., 2021)(Liu et al., 2025) . AdvFib PI16 hi states are mostly found in the fallopian tube ampulla and isthmus (99% of PI16 hi cells; Extended Data Fig. 3b), where spatial transcriptomics localise them to its characteristic subserosal (adventitial) connective tissue in the ampulla (Figure 2d). In contrast, AdvFib PI16low and Adv Fib (Intr) (i.e. C7/COL15A1 hi states) are distributed across tissues occupying the mucosal interstitium of the fallopian tube, the subserosa region beneath the ovarian surface epithelium and adventitial positions around vasculature in the ovary, fallopian tube and the myometrium in the uterus (Figure 2d and Extended Data Fig. 3c)."

(Liu et al., 2025)

[Fibroblast atlas: Shared and specific cell types across tissues]

"Understanding the heterogeneity of fibroblasts depends on decoding the complexity of cell subtypes, their origin, distribution, and interactions with other cells. Here, we integrated 249,156 fibroblasts from 73 studies across 10 tissues to present a single-cell atlas of fibroblasts. We provided a high-resolution classification of 18 fibroblast subtypes. In particular, we revealed a previously undescribed cell population, TSPAN8+ chromatin remodeling fibroblasts, characterized by high expression of genes with functions related to histone modification and chromatin remodeling. Moreover, TSPAN8+ chromatin remodeling fibroblasts were detectable in spatial transcriptome data and multiplexed immunofluorescence assays. Compared with other fibroblast subtypes, TSPAN8+ chromatin remodeling fibroblasts exhibited higher scores in cell differentiation and resident fibroblast, mainly interacting with endothelial cells and T cells through ligand VEGFA and receptor F2R, and their presence was associated with poor prognosis. Our analyses comprehensively defined the shared and specific characteristics of fibroblast subtypes across tissues and provided a user-friendly data portal, Fibroblast Atlas."